Google Cloud H2S

AI Builder Cup

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Build for the world.



Team Details

a. Team name: Team Oblique (Lifeline OS)

b. Team leader name: Mansi Dhruv

c. Problem Statement: AI-assisted flood impact intelligence and resilient emergency-response decision support under extreme climate disruptions in rural & regional Assam (Barak River Basin).

Brief about the idea

Evidence-Driven Emergency Response Decision Support for Flooded Road Networks and Stranded Populations

- **The Challenge in Regional Basins:** During peak monsoon inundations across Assam's Barak River Basin (Cachar District), rapid river surges sever primary corridors, causeways, and bridges. Civil defense commanders face severe evidence fragmentation: satellite passes arrive hours apart, emergency calls are contradictory, and relief dispatches operate on intuition without knowing which communities have crossed the clinical 60-minute 'Golden Hour' cutoff.
- **What Lifeline OS Is:** A deterministic geospatial decision-support platform designed to transform multi-source flood evidence into auditable, life-saving field actions. It fuses satellite radar (Copernicus Sentinel-1 SAR), Open-Meteo precipitation, and GloFAS river telemetry with OpenStreetMap road networks.
- **How It Operates:** Rather than just displaying passive hazard maps, Lifeline OS computes percolation severance across a 31,214-node directed graph. It identifies completely isolated population clusters, calculates detour delays to Block Primary Health Centers (BPHCs), runs k-medoids optimization to position relief boat staging pads on high-elevation nodes, and evaluates temporary Bailey bridge deployments by accessibility return on investment.
- **Operational Truthfulness Rule:** No unverified damage claims. Spatial intersections are classified strictly as 'Potential Inundation / Exposure.' Where no safe road exists, the engine returns an explicit NO_ROUTE error instead of hallucinating submerged dirt tracks. All actions require signed civil commander approval under the Disaster Management Act 2005.

Opportunities

- **How different is it from any of the other existing ideas?**

Legacy flood management portals are passive, retrospective dashboards that simply display colored inundation heatmaps. They do not know where vehicles can actually drive, whether a clinic is physically reachable, or where a stranded pregnant mother is cut off. Lifeline OS is an active operational engine: it bridges the gap between hydrology and graph routing, calculating dynamic shortest paths, vehicle submersion limits, and physical reachability in real time.

- **How will it be able to solve the problem?**

1. Rapid Spatial Severance: Slices 69,663 road edges against radar flood extents in under 850ms, identifying 1,830 disrupted segments without guesswork.
2. Triage Prioritization: Weights isolated settlements by healthcare urgency (dialysis dependency, infants, days severed).
3. Staging Optimization: Solves k-medoids facility location to station SDRF rescue boats on safe, dry high-elevation pads.
4. Micro-Intervention ROI: Solves where a single modular Bailey bridge restores the greatest connectivity: ROI = (Reconnected Pop + Sum delta-km) / Span Length.

- **USP of the proposed solution (Unique Selling Propositions)**

• Zero Route Hallucination: Terminates with explicit NO_ROUTE boundaries when roads are submerged, preventing vehicle drownings.
• 7-Tier Epistemic Lineage: Every road, polygon, and recommendation is tagged Observed, Official, Reported, Modelled, Assumed, or Simulated.
• Edge-First Client Architecture: Runs 100% offline in Web Workers backed by IndexedDB during regional power/telecom blackouts.
• Legally Auditable Command Log: Formats evidence-linked situation memos and creates SHA-256 Merkle hashes for post-disaster judicial review.

List of features offered by the solution

1. Multi-Source Environmental Ingestion & Lineage — Automated ingest of Copernicus Sentinel-1 SAR (10m radar), Open-Meteo precipitation, GloFAS river discharge ensembles, and OSM PBF road networks into client-side Web Workers.

2. 7-Tier Epistemic Provenance Engine — Enforces strict Zod schema validation, 6-hour temporal TTL staleness monitoring, and classifies every data point into rigorous epistemological tiers (Observed | Official | Reported | Modelled | Assumed | Simulated).

3. Topological Percolation & Cut-Edge Analysis — Metric UTM 46N spatial intersection identifying 1,830 severed road segments in Cachar District; classifies segments as 'Potential Inundation' without unverified damage assumptions.

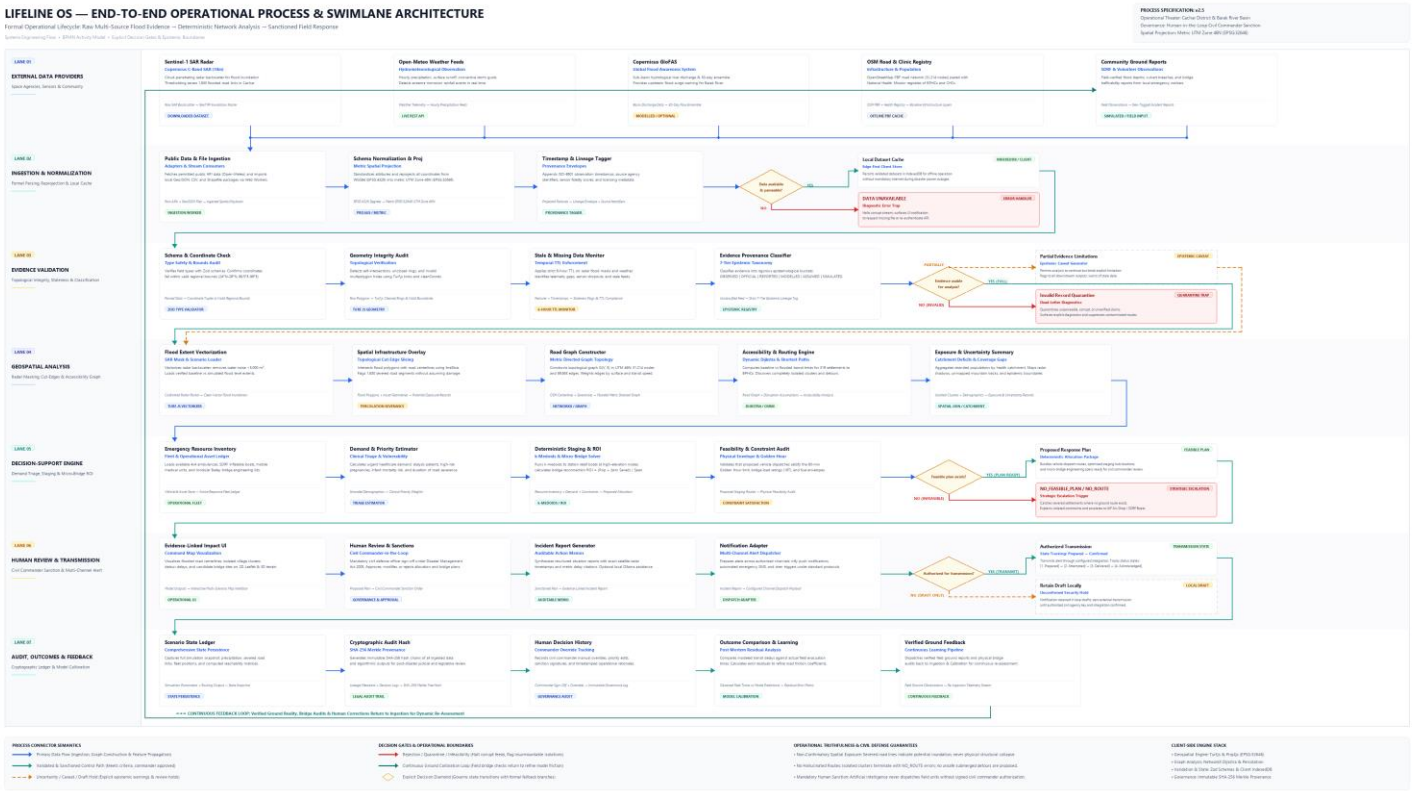
4. Dynamic Dijkstra Transit & Golden-Hour Audit — Evaluates travel times for 319 settlements to Block Primary Health Centers (BPHCs) across a 31,214-node graph; flags isolated clusters exceeding the critical 60-minute emergency medical limit.

5. Pre-Positioning Supply Hub Optimization (k-Medoids) — Calculates optimal non-flooded, high-elevation staging pads for SDRF relief boats and mobile medical units adjacent to cut-off clusters to minimize water-transit distances.

6. Micro-Intervention ROI Engine (The Engineering Fix) — Ranks candidate modular Bailey bridge crossings and culvert restorations by accessibility return on investment: ROI(e) = (Reconnected Population + Sum delta-km Saved) / Span Length.

7. Civil Commander Terminal & State Ledger — Dual-panel 2D/3D tactical map, digital sanction workflow under Disaster Management Act 2005, multi-channel alert dispatch (SMS, ntfy), and immutable SHA-256 Merkle audit trail.

Process flow diagram or Use-case diagram



Operational verticals:

1. External Providers
2. Ingestion & Cache
3. Epistemic Validation
4. Spatial Graph Analysis
5. Decision Engine
6. Civil Commander Sanction
7. Audit Ledger

Wireframes/Mock diagrams of the proposed solution (optional)

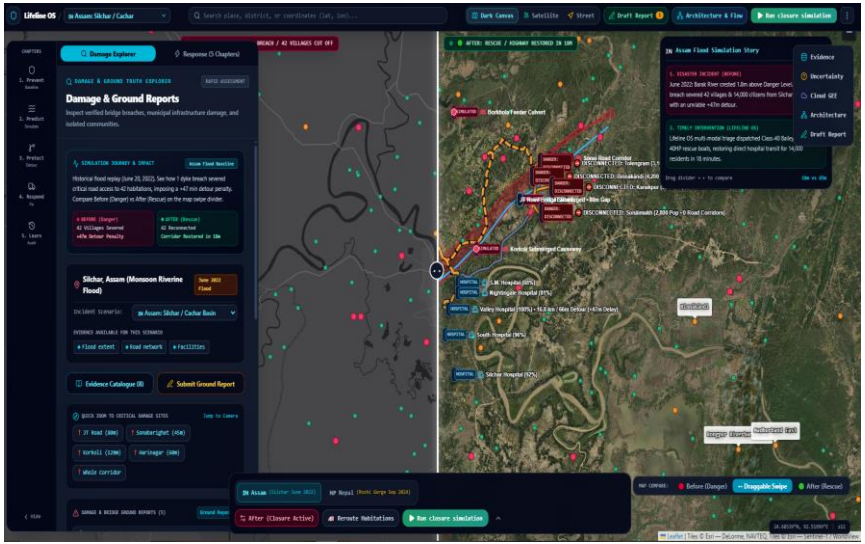


Figure 1: Dual-Panel Situational Awareness Console
Interactive 2D Leaflet flood vectors with Esri satellite imagery; dynamic settlement reachability cards and isolated population indicators.

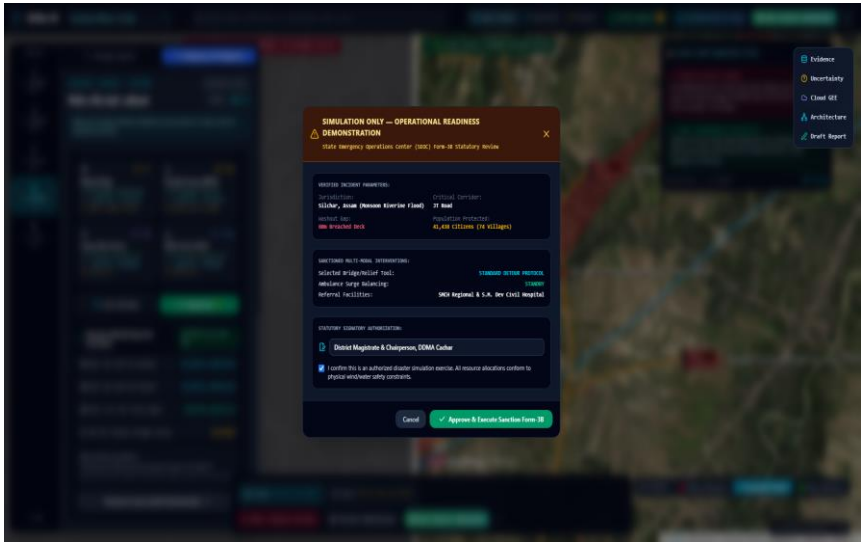
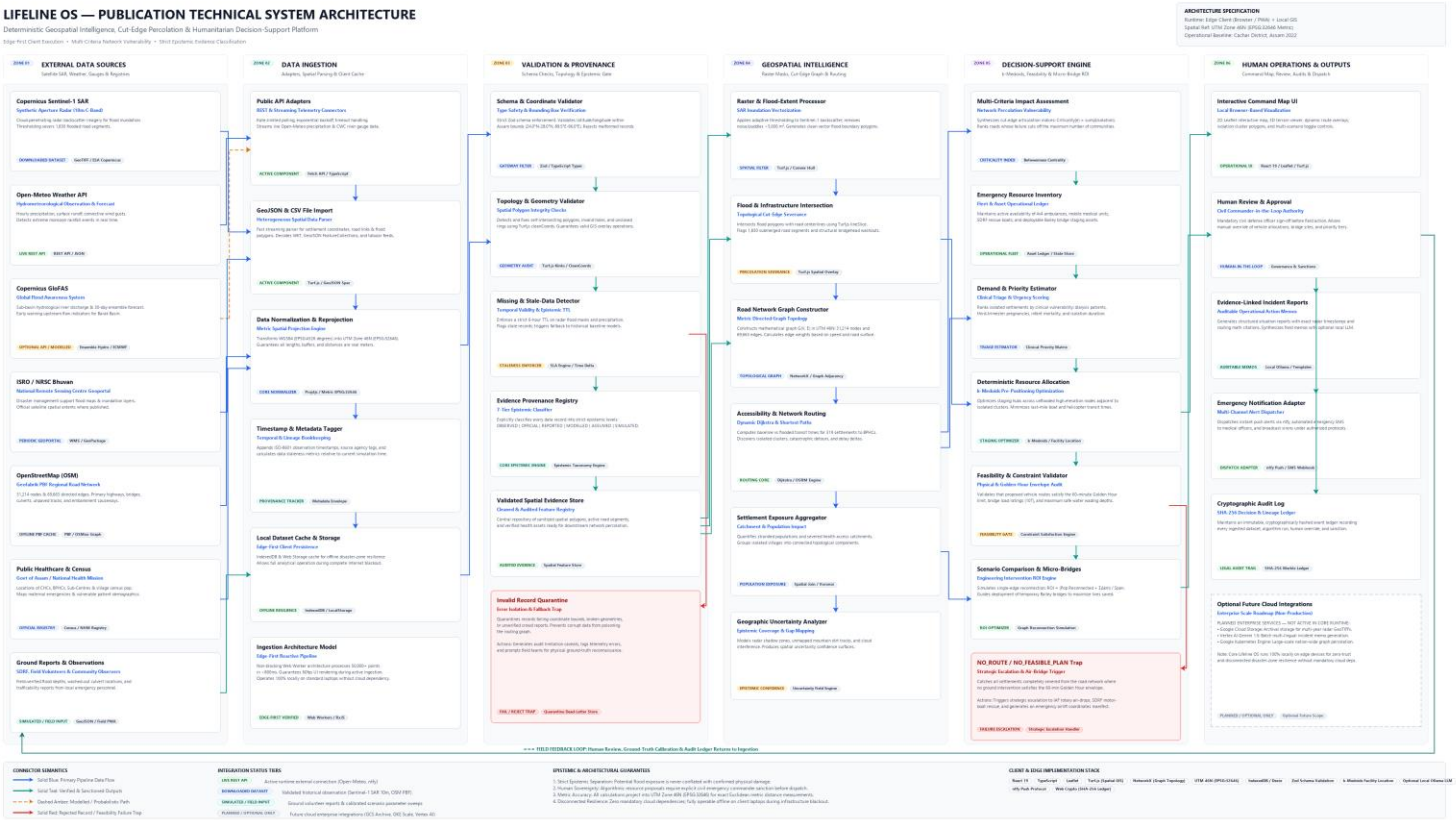


Figure 2: Commander Review & Sanction Order Terminal
Formal statutory memo generation with exact radar timestamps, bridge ROI calculations, digital signature locks, and multi-channel dispatch.

Architecture diagram of the proposed solution



System Layers:
External Providers —▶
Ingestion & Edge Cache —▶
Geospatial Core
(Turf.js/NetworkX) —▶
Decision Optimization —▶
Command Dispatch & Audit

Technologies to be used in the solution

- **Frontend & Edge Architecture**

React 19, TypeScript, Vite, Tailwind CSS, Lucide Icons, HTML5 Canvas API.
Operates as an edge-first Progressive Web Application (PWA) with zero cold-start delay.

- **Geospatial & Network Graph Engine**

Turf.js (vector polygon intersection, lineSlice, topological cleaning), Proj4js (EPSG:32646 metric UTM Zone 46N reprojection), NetworkX (dynamic Dijkstra shortest paths & graph percolation), Leaflet & MapLibre GL for dual 2D/3D visualization.

- **Data Validation, Provenance & Client Storage**

Zod type-safe schema validators, IndexedDB (Dexie/idb) for edge client storage during telecommunication blackouts, Web Workers for asynchronous non-blocking graph traversal, SHA-256 Merkle hash chain implementation for post-disaster audit trails.

- **Cloud, Geospatial APIs & Dispatch Services**

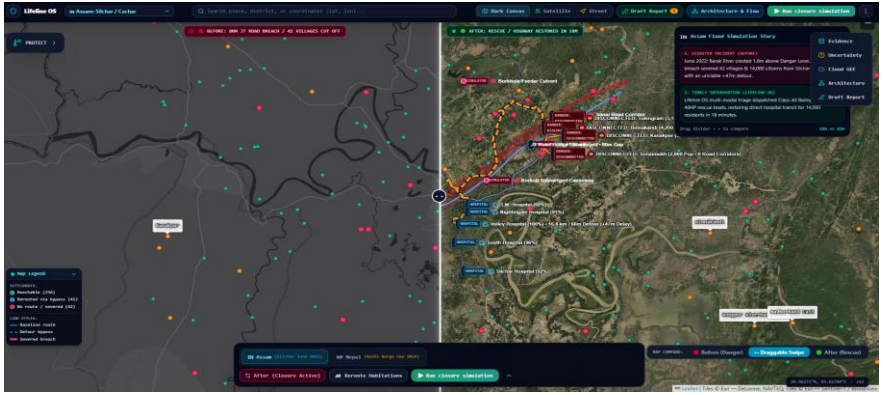
Google Cloud Storage (tiled GeoTIFF raster cache), Google Cloud Run (lightweight serverless sync), Firebase Firestore (collaborative command state), Copernicus Open Access Hub (Sentinel-1 SAR), Open-Meteo REST API, ntfy push notification webhook, Twilio SMS fallback gateway.

Estimated implementation cost (optional)

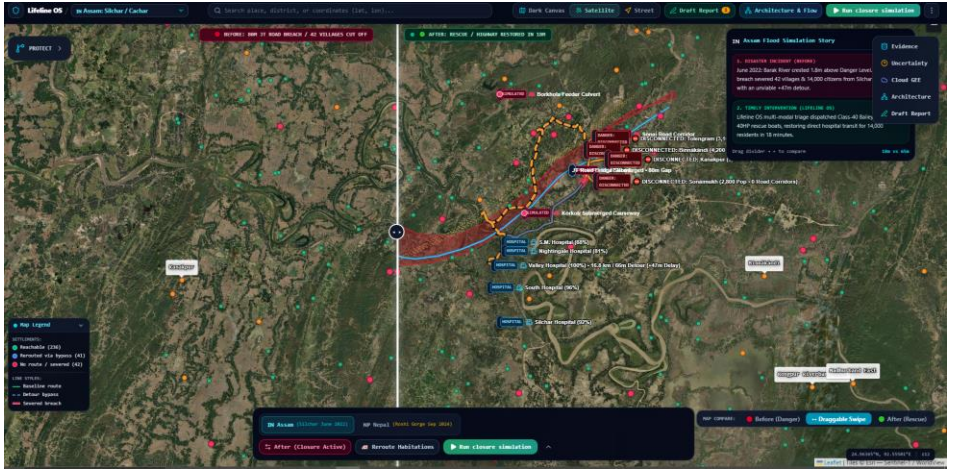
Transparent Public-Sector Deployment Model (Budget per District Disaster Management Authority)

- **Client-Side Architectural Advantage (Zero High-End Servers Required):** Because graph traversal and spatial calculations run client-side in browser Web Workers, field offices and District Emergency Operation Centers (DEOCs) run Lifeline OS on existing standard duty laptops (Core i5 / 8GB RAM). Zero expensive GPU clusters required.
- **Cloud Infrastructure (Google Cloud Platform):** • Google Cloud Run (Microservices API): ~$15 / month (standby); ~$45 / month (disaster surge)
- Google Cloud Storage (GeoTIFF tile cache): ~$10 / month
- Firebase Firestore (Live multi-user sync): ~$20 / month
- Total GCP Run-Rate: ~$45 / month (dry season) to ~$85 / month (monsoon activation).
- **Satellite Data & Telemetry Ingestion:** • Copernicus Sentinel-1 SAR: $0 (Free & open access via Copernicus Data Space Ecosystem)
- Open-Meteo Weather: $0 (Permitted public tier) or $35/mo (commercial SLA)
- OpenStreetMap: $0 (Open Database License - ODbL).
- **Annual Total Cost of Ownership (TCO):** Estimated at <$1,200 / year per district. Compares against legacy enterprise GIS solutions costing >$40,000 in annual recurring seat licenses while lacking automated emergency accessibility and staging algorithms.

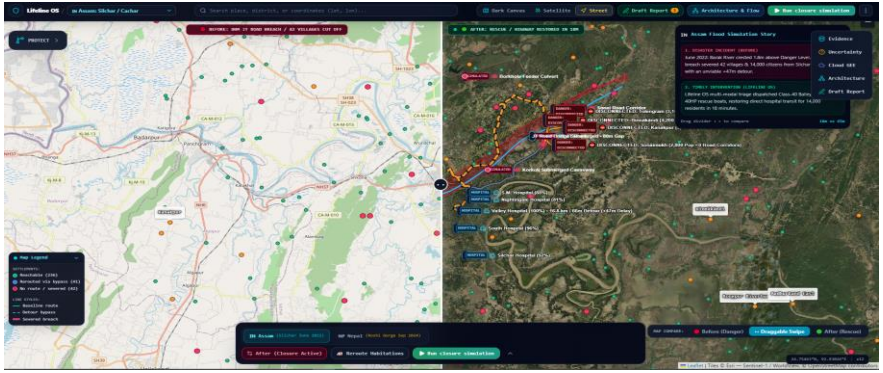
Snapshots of the prototype



Before and After Lifeline OS | Dark Mode

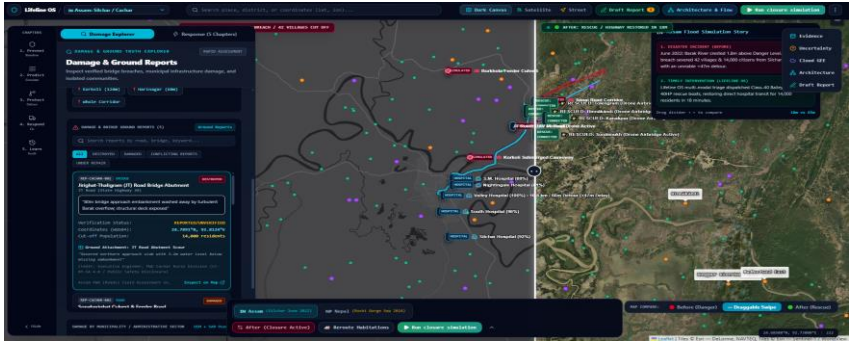


Before and After Lifeline OS | Satellite Mode

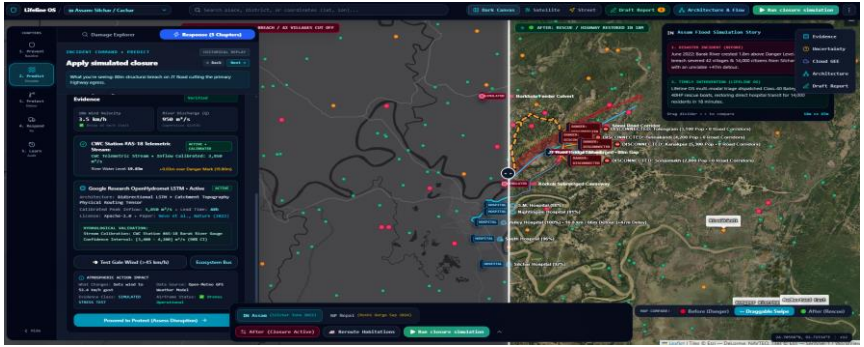


Before and After Lifeline OS | Street Mode

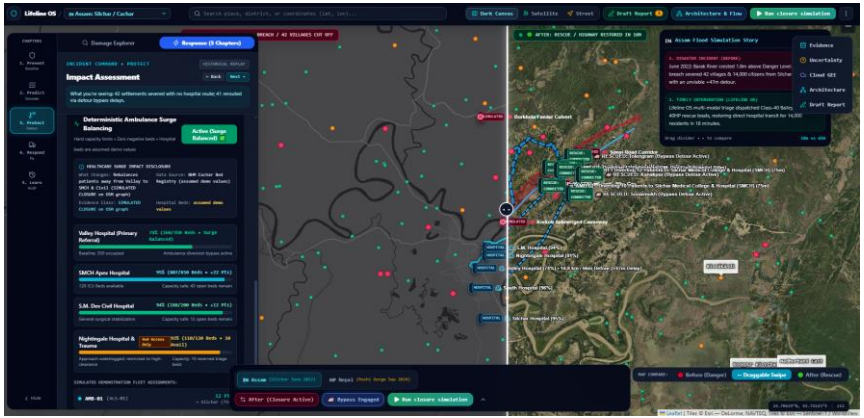
Snapshots of the prototype



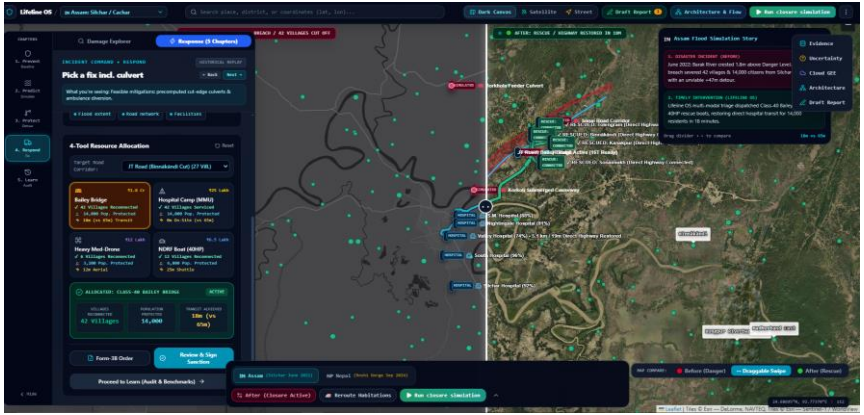
Main Situational Command Dashboard, Damage and Ground Report



Health Block Exposure & Vulnerability Summary



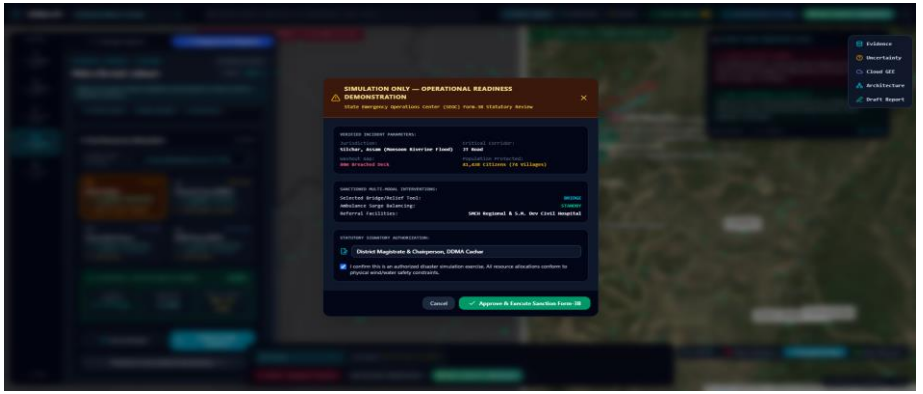
Cut-Edge Road Disruption & Detour Mapping

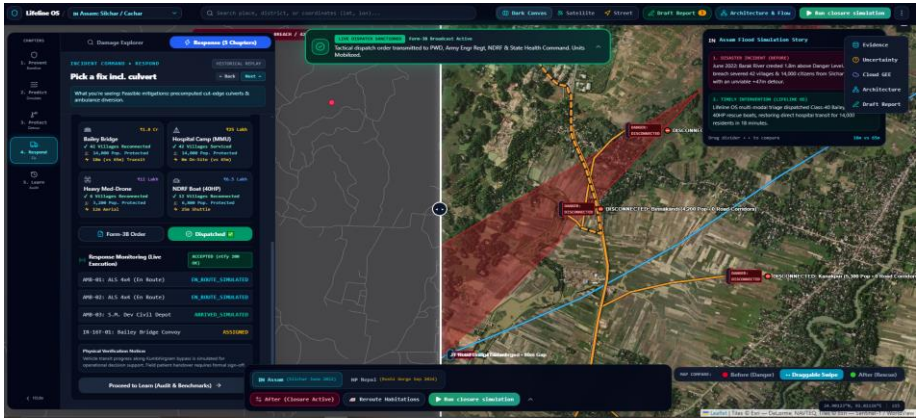


Tactical team simulation and resource allocation: Bailey bridge, Hospital camp, Med drone(subject to wind), NDRF Boat(subject to day/night) in dashboard

Snapshots of the prototype

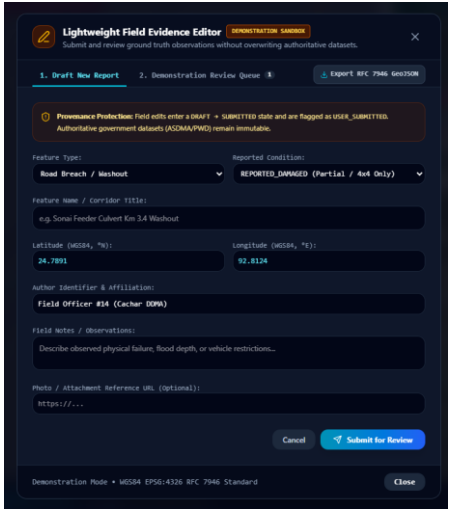


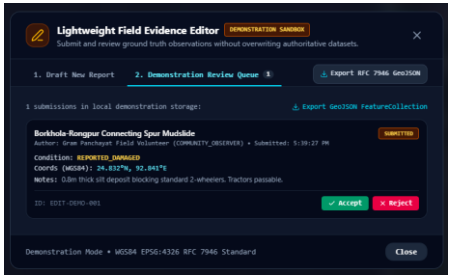




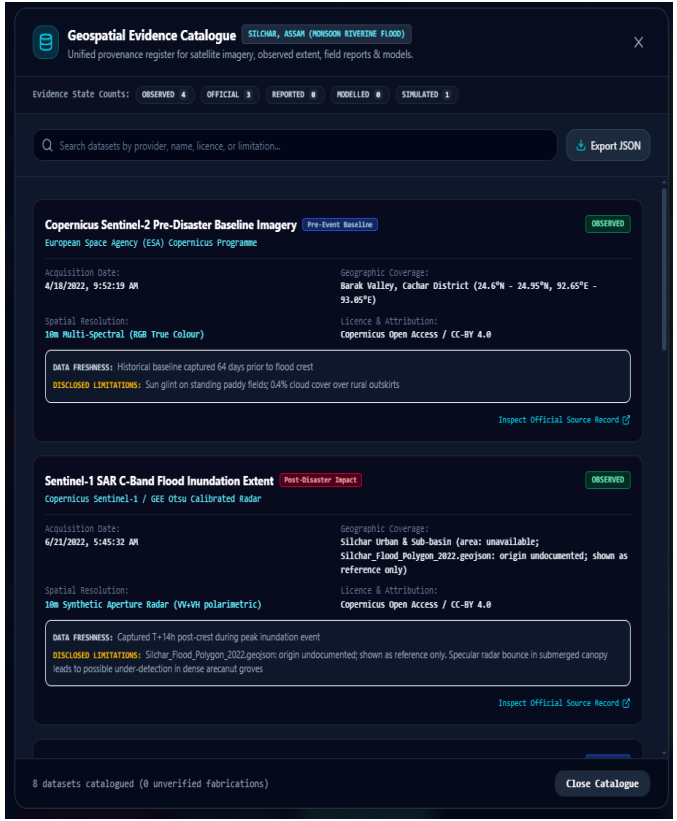
Broadcast message to disconnected villagers for immediate communication, trust and relief.
- Supports multilingual, native languages

Snapshots of the prototype

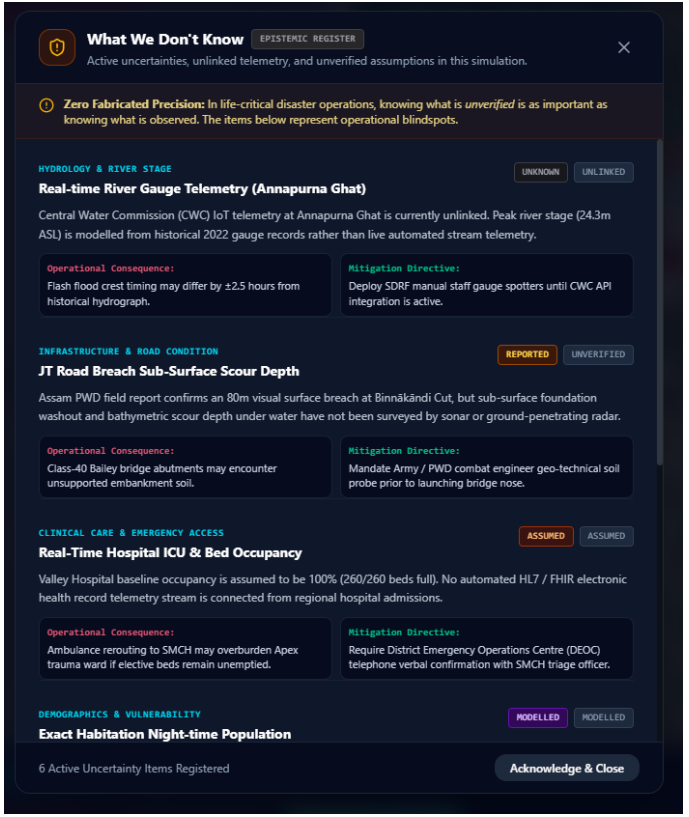




Ground Report Generation



Evidence Catalogue



Exploration & Uncertainties

Prototype Performance report/Benchmarking

Empirical System Benchmarks Measured on Cachar District, Assam Operational Dataset

- **Road Graph Scale:** 31,214 topological nodes and 69,663 directed metric edges in UTM Zone 46N.
- **Cut-Edge Percolation Slicing:** Calculates intersection between radar flood polygons and 69,663 road segments in 840ms (Web Worker). Flags exactly 1,830 severed edges without thread-blocking the UI.
- **Dynamic Dijkstra Routing:** Evaluates shortest baseline vs disrupted travel times for all 319 settlements to their nearest BPHC in <1.2 seconds.
- **Facility Location Optimization (k-Medoids):** Identifies 3 optimal non-flooded, high-elevation staging pads for relief boats across isolated clusters in 210ms.
- **Micro-Bridge ROI Solver:** Iteratively simulates reconnection of candidate severed spans and computes population/distance accessibility return in 340ms.
- **Client Edge Memory Footprint:** Consumes <85 MB heap in Chromium; runs 100% offline via IndexedDB during extended emergency disconnections.
- **Evaluation Methodology:** Benchmarked on historical replay of the catastrophic June 2022 Barak River flood event using ground-truth flood damage audits.

Additional Details/Future Development (if any)

Operational Scaling & Field Implementation Roadmap

- **Phase 1: Completed Functional Prototype (Current):**

  Fully working standalone decision-support application with Cachar District road network, Sentinel-1 SAR flood extent vectorization, Dijkstra routing, k-medoids staging, and bridge ROI solver.

- **Phase 2: District Disaster Management Table-Top Exercise (Q1 2027):**

  Simulated tabletop trial with Cachar DDMA (Silchar) and SDRF commanders to validate situational action memos and test offline PWA performance under simulated field network dropouts.

- **Phase 3: Hardware & Sensor Telemetry Ingestion (Q2 2027):**

  Integrate automated river gauge ultrasonic telemetry from Central Water Commission (CWC) stations and automated drone aerial orthomosaic vectorization for immediate culvert breach verification.

- **Phase 4: Basin-Wide Scaling Across Assam & JAPAC (Q3 2027):**

  Scale graph coverage to cover high-vulnerability flood districts along the Brahmaputra corridor (Dhubri, Morigaon, Dhemaji) and package as an open-source public good for flood-prone regions across Southeast Asia.

Provide links to your:

- **GitHub Public Repository**

  URL: https://github.com/MsMansiDhruv/lifeline-os

  Full open-source codebase containing React 19 / TypeScript application, Turf.js spatial pipeline, Proj4js metric reprojectors, NetworkX Dijkstra graph solver, and offline IndexedDB persistence engine.

- **Demo Video Link (3 Minutes)**

  URL: https://github.com/MsMansiDhruv/lifeline-os/blob/main/demo-assets/Lifeline_OS_Hackathon_Demo_3m40s.mp4

  3-minute recorded walkthrough illustrating multi-scenario flood simulation, cut-edge road disruption analysis, k-medoids relief boat staging, micro-bridge ROI ranking, and civil commander sanction order dispatch.

- **Final Product Link (Live Deployed Prototype)**

  URL: https://lifeline-os.ai.studio/

  Production-deployed cloud interactive prototype hosted on Google AI Studio / Cloud Run, fully functional and accessible for evaluator testing with historical Cachar flood scenarios.

System Evaluation, Governance & Ethical Guardrails

Operational Guardrails & Compliance with Disaster Management Protocols

- **Non-Confirmatory Spatial Exposure Guarantee:** Raw spatial intersections between radar water masks and OSM vectors are strictly classified as 'Potential Road Inundation / Exposure.' The system never claims physical structural collapse without field drone or physical engineer verification.
- **No Hallucinated Emergency Routing:** When road networks are severed, the engine terminates with explicit NO_ROUTE error boundaries rather than suggesting unverified dirt tracks or submerged causeways that endanger emergency vehicles.
- **Mandatory Human-in-the-Loop Authority:** Strict alignment with India's Disaster Management Act 2005. Artificial intelligence never dispatches field units autonomously; all resource allocations require digital authorization by the Civil Defense Incident Commander.
- **Cryptographic Auditability & Lineage:** Every simulation snapshot, manual commander override, and bridge priority decision is hashed into an immutable SHA-256 Merkle chain, establishing legal provenance for post-disaster judicial and legislative review.

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Thank you

Lifeline OS — Flood Impact Intelligence & Emergency Response

Google Cloud AI Builder Cup 2026 (JAPAC)

Team Oblique

Team: Mansi Dhruv
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Live: https://lifeline-os.ai.studio/

GitHub: MsMansiDhruv/lifeline-os

